

Shared Memory and Memory-Mapped Files

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Definition

Inter Process Communication (IPC) is a mechanism that allows two or more processes to communicate and exchange data with each other. It helps processes share information and coordinate their activities while running on the same computer or across different systems.

Shared Memory and Memory-Mapped Files

Shared Memory is an IPC technique where two or more processes share a common area of memory. Processes can directly read from and write to this shared memory, making communication very fast.

A Memory-Mapped File maps a file or part of a file directly into a process's memory. This allows processes to access the file's contents as if they were working with data in memory, making file sharing and communication more efficient.

Example

Consider two processes in a banking application. One process updates a customer's account balance, while another process needs to display the updated balance.

The processes can use shared memory to store the account balance. The first process writes the updated balance into the shared memory, and the second process reads it directly.

For a memory-mapped file, the account information can be stored in a file that is mapped into memory. Both processes can access the mapped data without repeatedly reading and writing the entire file.

Process 1 ↔ Shared Memory / Memory-Mapped File ↔ Process 2